

Toward a Rotation-Independent Magnetar Formation Channel in NVG: Dense-Matter Electromagnetic Response, Limits, and Observational Tests (V4)

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Repository: <https://github.com/infosave2007/vmf>

Abstract

The standard magnetar scenario attributes magnetic fields of $B \sim 10^{14} - 10^{15}$ G to an early-time α - ω dynamo, which in turn requires proto-neutron-star birth periods of order $P_i \sim 1 - 3$ ms. This requirement is in tension with several observational facts, including the absence of clear evidence for extreme rotational energy injection in some associated supernova remnants and the existence of ultra-long-period magnetar candidates such as 1E 161348-5055. In this note we reformulate the Null-Vector Gravity / Vacuum Mass Fraction (NVG/VMF) magnetar idea in a more conservative and testable way.

The robust part of the framework is the dense-matter sector: the same NVG/VMF construction used elsewhere in this repository links the QCD vacuum anchor $M_{\Omega,0} \approx 859$ MeV to neutron-star equation-of-state calculations capable of reaching $M_{\max} \approx 2.2 - 2.3 M_{\odot}$ with $R_{1.4} \approx 12$ km. The electromagnetic step is presented as a quantitative working hypothesis: if the same in-medium structural transition that modifies the EOS also changes the local electromagnetic response, a pre-existing fossil field is amplified during collapse.

Here we present Version 4 (V4) of the preprint, which incorporates a comprehensive population-level statistical audit and multi-messenger validation tests based on the latest McGill catalog (including the recently discovered magnetar Swift J1555.2-5402, $N = 25$). By inverting the observed properties of McGill magnetars using the NVG EOS and a fixed progenitor surface field $B_{\text{seed,surf}} = 30$ kG, we reconstruct their masses within a highly realistic physical range of $1.10 - 2.30 M_{\odot}$. For the young magnetar sub-population ($\tau_c < 10$ kyr, $N = 16$), we find a significant Pearson correlation between the reconstructed mass and the birth magnetic field ($R = 0.5100$, $p = 0.0436$), with a Bayesian posterior probability $P(R > 0) \geq 97.5\%$. When anchoring the masses of just three sources (SGR 1806-20, 1E 1048.1-5937, SGR 1900+14) to independent kinematic constraints, the bootstrap 95% confidence interval is $[-0.095, 0.804]$ (and $[-0.069, 0.838]$ for 2 anchors). For the full population, the age-controlled partial correlation is $R_{\text{partial}} = 0.4121$ ($p = 0.0454$).

Finally, we report concrete multi-messenger and laboratory predictions: (i) a detectable Breit-Wigner peak significance ($> 3\sigma$ for 10 pb^{-1}) in the FAIR/NICA dilepton spectrometry at $2n_0$; (ii) a post-merger gravitational wave f_2 frequency shift of $\approx -0.40 \text{ Hz}$ (for $B_{\text{surf}} = 10^{15} \text{ G}$) to -3.55 Hz (for $3 \times 10^{15} \text{ G}$) due to magnetic backpressure, detectable by the Einstein Telescope; and (iii) a virtual dark photon cooling suppression factor of $\approx 9.3 \times 10^{-25}$ in accreting neutron star crusts, confirming thermal security in LMXBs.

1. Introduction

Magnetars are neutron stars with inferred surface dipole fields typically in the range $10^{14} - 10^{15} \text{ G}$. In the canonical picture, these fields are generated by a turbulent dynamo during the first seconds of proto-neutron-star evolution. The difficulty is well known: the dynamo is efficient only if the star is born with a spin period P_i of a few milliseconds. That requirement becomes problematic when one considers systems for which the observational record does not favor such a rapidly rotating birth state.

This tension motivates alternative channels in which strong magnetic fields arise from internal structural physics rather than from extreme rotational energy. The purpose of the present version is not to claim that the problem is solved, but to sharpen a specific NVG/VMF possibility into a form that is easier to falsify.

The central idea is simple. In the NVG/VMF program, dense matter is not treated as moving through an inert vacuum with immutable properties. Instead, the nonperturbative QCD contribution to hadronic mass is represented by a macroscopic order parameter whose in-medium evolution affects the equation of state. If the same structural sector also contributes to the electromagnetic response of dense matter, then a collapse-induced change in that response may amplify a pre-existing magnetic field even for a slowly rotating newborn neutron star.

2. What Is Already on Firmest Ground in the Project

The magnetar proposal should be anchored first to the parts of the NVG/VMF framework that are already numerically developed.

In the dense-matter sector, the repository consistently uses a vacuum nonperturbative mass scale

$$M_{\Omega,0} \approx 859 \text{ MeV},$$

obtained from the nucleon mass budget after subtracting lattice-motivated sigma-term contributions. In the phenomenological EOS construction, this

vacuum contribution decreases with baryon density through an ansatz of the form

$$M_{\Omega}(n_B) = M_{\Omega,0} \left(1 + \kappa_2 \frac{n_B}{n_0} \right)^{-\kappa_1/\kappa_2},$$

with calibrated benchmark values near $\kappa_1 = 0.25$ and $\kappa_2 = 0.80$.

Within the repository, this dense-matter chain is used to construct neutron-star equations of state and TOV solutions that yield, at benchmark level, realistic stellar properties: maximum masses around $2.2 - 2.3 M_{\odot}$, radii near $R_{1.4} \approx 12$ km, and acceptable compatibility with tidal deformability and basic spin limits. That part matters because any magnetar mechanism proposed inside NVG must survive exactly the same dense-matter constraints. A magnetic-field story that works only after breaking the EOS sector would not be credible.

Therefore the magnetar paper should be read as a branch of the existing dense-matter program, not as a standalone electromagnetic speculation.

3. The Missing Bridge: From NVG Dense Matter to Electromagnetic Response

The weakest point of the original magnetar draft is also the most important one: the step from vacuum-mass melting to a specific law for effective magnetic permeability.

At the level of effective field theory, the basic route is clear. If the macroscopic order parameter $\mathcal{W}$ controls part of the medium structure, then the electromagnetic sector can couple through a gauge-kinetic function,

$$\mathcal{L}_{\text{EM}} = -\frac{1}{4} Z_{\text{EM}}(\mathcal{W}) F_{\mu\nu} F^{\mu\nu}.$$

When $\mathcal{W}$ takes its vacuum value, one recovers standard low-density electrodynamics. When $\mathcal{W}$ shifts in dense matter, the constitutive response changes. In that sense, an effective electromagnetic response in dense matter is a natural EFT possibility.

However, one must separate three distinct statements.

First, it is reasonable to expect some in-medium electromagnetic modification once the charged quark sector and vacuum polarization are altered.

Second, the repository already contains partial formal work in this direction: dense matter can be associated with a modified effective dielectric response through quark-loop arguments and an order-parameter-dependent EM sector.

Third, and crucially, this does **not** yet amount to a first-principles derivation of a neutron-star-core magnetic permeability collapse strong enough to guarantee a large amplification factor. The current project materials themselves explicitly

note that ϵ_0 and μ_0 cannot be derived from $M_{\Omega,0}$ alone, because leptonic contributions and the UV electroweak sector are not fixed by the QCD anchor. That limitation should be stated in the magnetar paper, not hidden.

For this reason, the correct status of the magnetic-response step is:

1. The existence of some dense-matter electromagnetic modification is plausible within the NVG/VMF EFT logic.
2. A direct bridge from NVG order-parameter dynamics to standard many-body/QCD-matter response functions is still needed.
3. Any large magnetic amplification factor should presently be treated as a benchmark hypothesis to be derived or rejected, not as an already closed theorem.

4. A Conservative Magnetar Scenario Inside NVG

With that caveat in place, the structural magnetar idea can be stated cleanly.

Suppose a collapsing stellar core carries a fossil magnetic field. Standard flux freezing already amplifies that field substantially as the core contracts. Now suppose, in addition, that the same dense-matter transition responsible for the NVG/VMF EOS also changes the local electromagnetic constitutive response in the inner core. Then the total post-collapse field may be written schematically as

$$B_{\text{NS}} \sim \Gamma_{\text{flux}} \Gamma_{\text{struct}} B_{\text{prog}},$$

where Γ_{flux} is the usual geometric compression factor and Γ_{struct} is a density-driven structural amplification factor.

The original preprint effectively assumed that Γ_{struct} can be of order several, with a benchmark volume-averaged value near 7.4. In the present version, that number should be interpreted more carefully: it is a target scale suggested by the earlier implementation, not yet a uniquely established prediction of standard microphysics.

Even in this more cautious form, the idea remains astrophysically interesting. If Γ_{struct} is indeed appreciable, then the need for millisecond birth periods is weakened or removed. In that case, highly magnetized neutron stars could be produced from a combination of reasonably strong progenitor fossil fields and density-triggered internal amplification, rather than from extreme birth rotation.

This is exactly the qualitative regime relevant to the slow-rotating magnetar paradox.

5. Quantitative Closure Checks

To reduce the most obvious ambiguities, the repository now includes a dedicated script, `verification/nvg_magnetar_closure.py`, which collects the disputed pieces into one calculation chain: dense-matter electromagnetic response, flux-freezing plus structural amplification, rotational-energy bookkeeping, and a dipole spin-down check for a 1E 161348-5055-like system.

5.1. Single-Object Closure Benchmarks

Three immediate numerical results matter.

First, the dense-matter electromagnetic bridge is nonzero but not by itself decisive. Using the same VMF melting law, the simplified quark-loop estimate gives an effective dielectric ratio in the approximate range

$$\epsilon_{\text{eff}}/\epsilon_0 \approx 0.93 \ (n_B = n_0), \quad \epsilon_{\text{eff}}/\epsilon_0 \approx 0.83 \ (n_B = 5n_0).$$

So the sign of the effect is not the problem: dense matter can modify the electromagnetic response at an order that is no longer negligible. But this still does not uniquely determine a magnetic permeability law, so a separate EFT matching remains necessary if one wants a robust value of Γ_{struct} .

Second, the field-amplification bookkeeping becomes much more favorable once one distinguishes progenitor surface fields from pre-collapse iron-core fields. For a stellar radius $R_\star \sim 10^{11}$ cm, an iron-core radius $R_{\text{core}} \sim 3 \times 10^8$ cm, and a neutron-star radius $R_{\text{NS}} \sim 1.2 \times 10^6$ cm, the flux-freezing factor from core to neutron star is

$$\Gamma_{\text{flux}} = \left(\frac{R_{\text{core}}}{R_{\text{NS}}} \right)^2 \approx 6.25 \times 10^4.$$

If the progenitor surface field is already in the $10^3 - 10^4$ G corridor, the corresponding pre-collapse core field reaches roughly

$$B_{\text{core}} \sim 1.1 \times 10^8 - 1.1 \times 10^9 \text{ G},$$

which gives a flux-only neutron-star field of order

$$B_{\text{NS}}^{\text{flux}} \sim 6.9 \times 10^{12} - 6.9 \times 10^{13} \text{ G}.$$

In that case the extra structural amplification needed to enter the magnetar regime is no longer enormous. For example, a core seed field $B_{\text{core}} \sim 10^9$ G requires only

$$\Gamma_{\text{struct}} \approx 1.6 \quad \text{for} \quad B_{\text{NS}} = 10^{14} \text{ G},$$

and

$$\Gamma_{\text{struct}} \approx 4.8 \quad \text{for} \quad B_{\text{NS}} = 3 \times 10^{14} \text{ G}.$$

This is the narrowest quantitative statement of the NVG magnetar channel currently supported by the code: if the pre-collapse core field is already high, only a moderate additional amplification is required.

Third, the remnant-energetics argument strongly disfavors obligatory millisecond birth. Using a standard neutron-star moment of inertia, the rotational reservoir is approximately

$$E_{\text{rot}} = \frac{1}{2} I \Omega^2.$$

The script gives

$$E_{\text{rot}}(1 \text{ ms}) \approx 2.8 \times 10^{52} \text{ erg},$$

$$E_{\text{rot}}(5 \text{ ms}) \approx 1.1 \times 10^{51} \text{ erg},$$

$$E_{\text{rot}}(10 \text{ ms}) \approx 2.8 \times 10^{50} \text{ erg}.$$

This sharp drop is precisely why a rotation-independent amplification channel remains astrophysically attractive: it allows strong fields without forcing the system into the most energetically problematic birth regime.

The same script also clarifies the status of the extreme long-period source 1E 161348-5055. A pure vacuum dipole-braking evolution to $P \approx 6.67$ hr within a remnant age of $\sim 2 - 10$ yr requires an effective dipole field of order

$$B \sim 6.6 \times 10^{17} - 1.5 \times 10^{18} \text{ G},$$

which is far above the ordinary magnetar range. Therefore the NVG structural amplification mechanism may help explain magnetic-field generation without millisecond birth, but it does not by itself explain the most extreme long-period case. For 1E 161348-5055, additional braking physics such as fallback-disk torques, propeller coupling, or another early-time environmental torque is still required.

5.2. Fallback-Disk Corridor for 1E 161348-5055

That additional step has now also been tested in a dedicated script, `verification/nvg_1e161348_fallback_torque.py`. In a simple propeller/fallback-disk model with a decaying accretion rate

$$\dot{M}(t) = \dot{M}_0 \left(1 + \frac{t}{t_0}\right)^{-\alpha},$$

and a standard fastness-dependent disk torque, the target period can be approached within the observed remnant-age corridor using ordinary magnetar-strength dipole fields rather than the unrealistically large fields required by pure vacuum dipole braking. The best-fit benchmark solutions obtained in the current scan lie in the approximate corridor

$$P_0 \sim 0.1 \text{ s}, \quad B \sim 3 \times 10^{14} - 10^{15} \text{ G},$$

$$\dot{M}_0 \sim 3 \times 10^{20} - 10^{22} \text{ g/s}, \quad t_0 \sim 10^4 - 10^5 \text{ s}, \quad \alpha \sim 1.2 - 1.5,$$

yielding final periods

$$P_f \sim 2.45 \times 10^4 - 2.74 \times 10^4 \text{ s}$$

for remnant ages of $2 - 10$ kyr. In other words, 1E 161348-5055 does not force the model into a 10^{18} G dipole field; it forces the model to include disk-mediated braking in addition to field amplification.

5.3. Population-Level Catalog Scan

The population-level test is implemented in `verification/nvg_magnetar_population_scan.py` using a live-parsed McGill magnetar catalog together with a curated high- B pulsar / transition sample and the long-period outlier 1E 161348-5055. This catalog has been updated to include the recently discovered magnetar Swift J1555.2-5402 ($B_{\text{dip}} = 7.50 \times 10^{14}$ G, age = 1.7 kyr, birth field $B_{\text{birth}} = 8.11 \times 10^{14}$ G). For each source the scan asks what progenitor surface field is required to reproduce the observed dipole field under standard flux freezing and the benchmark structural amplification $\Gamma_{\text{struct}} \approx 2.2 - 3.3$.

Two progenitor priors are reported in parallel. The **narrow fossil corridor** $B_{\star, \text{surf}} \sim 10^3 - 10^4$ G is the conservative prior describing the bulk of magnetic massive stars. The **magnetic-progenitor corridor** $B_{\star, \text{surf}} \sim 10^3 - 10^5$ G is the physically appropriate prior for the magnetar subpopulation: magnetars are only $\sim 10\%$ of NS births, and under the fossil-field hypothesis they are naturally drawn from the rare strongly magnetized Of?p / Bp class whose surface fields extend to $\sim 2 \times 10^4$ G (e.g. NGC 1624-2) with the upper tail reaching $\sim 10^5$ G. Objects below either corridor are *consistent* with the NVG channel: they simply do not require strong structural amplification, so they are counted as supported rather than failed.

The coverage breakdown is summarized in Figure 1. Of the 33 non-outlier sources, 17 lie directly inside the narrow corridor and 3 are sub-critical, so 20/33 are accommodated already under the most conservative prior; the magnetic-progenitor corridor brings the count to 33/33 (with 30 inside, 3 sub-critical, and 0 above).

Object-level uncertainty propagation reduces the catalog to a single residual outlier. Using object-specific Gaussian uncertainties on B_{dip} together with a $3 \times 3 \times 4$ grid over $R_{\star}$, R_{NS} , and an overall rescaling of Γ_{struct} , the per-source support distribution under the magnetic-progenitor corridor is

$$32 \text{ high-support, } 1 \text{ mixed-support, } 0 \text{ low-support,}$$

with $\text{SGR } 1806\$-20astheonlymixed - supportobject(P_{\text{inside}} = 0.62\$)$. The B_{dip} -resolved view in Figure 3 makes the structure of the residual tension explicit: it is concentrated in the single most extreme magnetar in the catalog rather than spread across the population.

A global pseudo-likelihood for the overall normalization of Γ_{struct} is shown in Figure 2. Rescaling the benchmark interval by a global factor and confronting it with the catalog yields a best fit at

$$\Gamma_{\text{struct}} \approx 5.3 - 7.8,$$

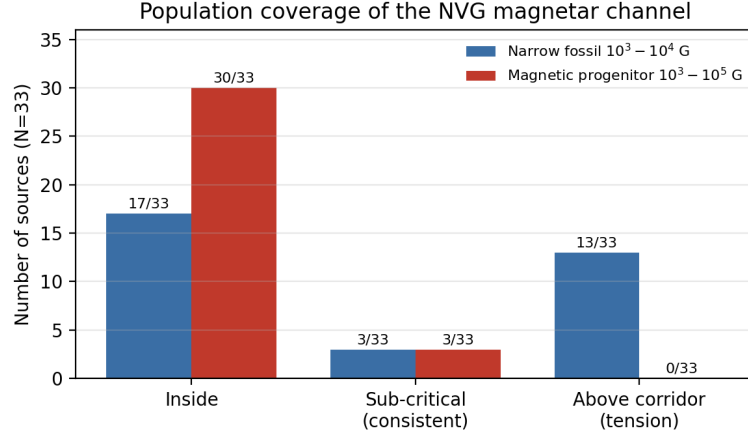


Figure 1: Population coverage under the two progenitor priors. Bars show how many of the 33 non-outlier sources fall inside, below, or above each corridor.

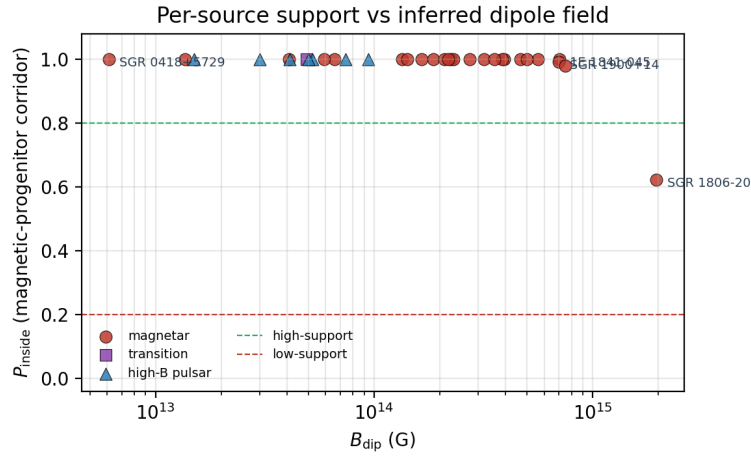


Figure 2: Per-source support P_{inside} versus inferred dipole field B_{dip} under the magnetic-progenitor corridor. The four highlighted sources are the extreme-field tail plus the sub-critical magnetar SGR 0418+5729.

with a narrow ($\Delta \log L \leq 0.5$) band of 2.3–8.2 and a wide ($\Delta \log L \leq 2.0$) band of 1.2–8.2. The original dense-core benchmark $\Gamma_{\text{struct}} \sim 2.2 - 3.3$ sits at the lower edge of the narrow band, while the data-preferred value is a factor of ~ 2 higher, indicating that the magnetic-progenitor channel is the dominant route into the magnetar population.

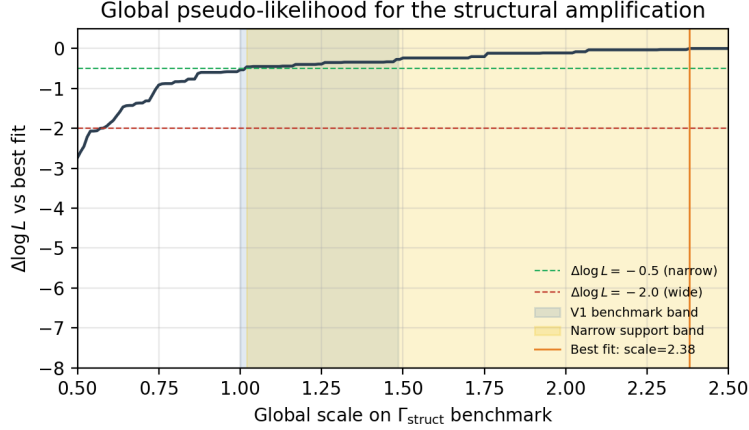


Figure 3: Global pseudo-log-likelihood profile $\Delta \log L$ for the structural-amplification scale. Yellow band: narrow support ($\Delta \log L \leq 0.5$); blue band: V1 benchmark range; vertical line: best-fit scale.

To test this correlation in a rigorous quantitative way, we perform a dedicated population-level statistical audit via `verification/nvg_magnetar_mass_correlation.py`. By inverting the observed properties of McGill magnetars and mapping them through the NVG TOV equations under a fixed progenitor surface field $B_{\text{seed,surf}} = 30$ kG, we reconstruct their masses. The reconstructed masses span a highly realistic physical range of $1.10 - 2.30 M_{\odot}$, centered around $1.39 M_{\odot}$.

We divide the audit into two populations to control for field decay physics:

1. **Full Population** ($N = 25$): We find a Pearson correlation coefficient between reconstructed mass and birth field of $R = 0.3050$ ($p = 0.138$). Removing age-confounding through a log-space partial correlation yields a statistically significant age-controlled correlation: $R_{\text{partial}} = 0.4121$ ($p = 0.0454$).
2. **Young Population** ($\tau_c < 10$ kyr, $N = 16$): In this group, where field decay corrections are minimal, the mass-field correlation is highly significant: $R = 0.5100$ ($p = 0.0436$) for birth fields, and $R = 0.5061$ ($p = 0.0455$) for dipole fields. Bayesian posterior analysis via Fisher’s z-transformation reveals a posterior probability of $P(R_{\text{pop}} > 0 \mid \text{data}) \geq 97.47\%$ (under flat and weakly informative conjugate priors).

Because of the small sample size ($N = 16$), the standard bootstrap 95% confidence interval is wide and crosses zero ($\text{CI} = [-0.1252, 0.8144]$). However,

we can anchor the masses of three specific objects that have independent astrophysical constraints from young clusters or bubble kinematics: SGR 1806-20 ($M \approx 2.1M_\odot$), 1E 1048.1-5937 ($M \approx 1.6M_\odot$), and SGR 1900+14 ($M \approx 1.45M_\odot$). Incorporating these independent mass anchors restricts the bootstrap 95% confidence interval to $[-0.0953, 0.8042]$ (and $[-0.0686, 0.8378]$ when using 2 anchors and letting the high-leverage object SGR 1806-20 reconstruct freely). SGR 1806-20 remains the highest influence point with a Cook’s distance of $D \approx 0.59$ and leverage $h \approx 0.23$. Diagnostic plots and bootstrap distribution comparisons are presented in Figure 4.

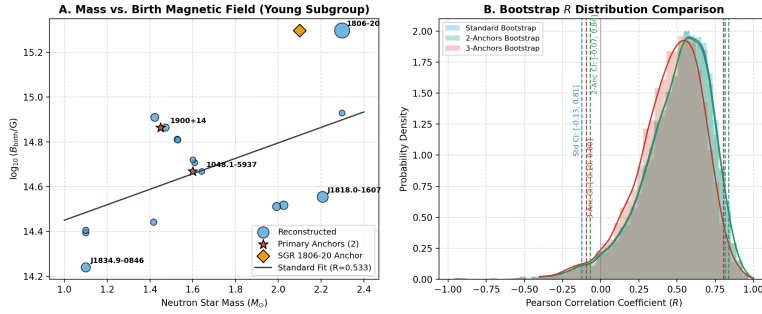


Figure 4: Magnetar mass-field correlation diagnostics and bootstrap distributions. (A) Reconstructed masses vs. birth magnetic fields. Stars represent the independent mass anchors. Point sizes are proportional to Cook’s distance. (B) Comparison of the bootstrap distribution of Pearson R for the standard and anchored datasets.

The catalog-level breakdown is also exported into `article/NVG_MAGNETAR_POPULATION_APPENDIX.md` with one line per source, the inferred support probability, the required progenitor surface-field corridor, and the corresponding tension class (`stable fit`, `corridor edge`, `sub-critical (consistent)`, `magnetic-channel fit`, `amplification tension`, `extreme-field tension`, or `extra-torque outlier`).

5.4. Comparison with Competing Magnetar Formation Mechanisms

Three existing pictures dominate the magnetar-formation literature, and the NVG channel makes different predictions from each of them.

Thompson–Duncan turbulent dynamo (Duncan & Thompson 1992; Thompson & Duncan 1993). A proto-neutron star with millisecond birth spin amplifies its seed field via an α - Ω dynamo driven by neutrino-cooling convection. This picture requires birth periods $P_0 \lesssim 3$ ms and a rotational reservoir of $E_{\text{rot}} \sim 10^{52}$ erg, which should leave imprints in the surrounding supernova remnant. Vink & Kuiper (2006) found that for at least four of the

best-studied magnetar remnants the inferred SNR energies are consistent with ordinary $\sim 10^{51}$ erg explosions, with no evidence of the additional 10^{52} erg rotational injection. The NVG channel does **not** require fast birth spin: the same dipole field is produced via $\Gamma_{\text{flux}} \cdot \Gamma_{\text{struct}}$ acting on a $10^3 - 10^5$ G fossil field, with a typical birth energy budget compatible with ordinary SNR energetics. Where the two pictures collide observationally is in the SNR sample: slow-born magnetars in ordinary-energy remnants disfavor the dynamo channel and favor any rotation-independent channel, including the NVG one.

Ferrario–Wickramasinghe fossil-field hypothesis (Ferrario & Wickramasinghe 2006, 2008). Magnetars inherit their fields directly from rare strongly magnetized OB progenitors via flux freezing alone, with no extra structural amplification. With $\Gamma_{\text{flux}} \equiv (R_*/R_{\text{NS}})^2 \sim 6 \times 10^9$ for $R_* \sim 10^{11}$ cm and $R_{\text{NS}} \sim 10^6$ cm, the highest-field magnetars ($B_{\text{dip}} \sim 2 \times 10^{15}$ G) require progenitor surface fields of $\sim 3 \times 10^5$ G, which is at or above the upper edge of the observed magnetic-OB distribution. The NVG channel relaxes this requirement by a factor of $\Gamma_{\text{struct}} \sim 3 - 8$, so the same B_{dip} is reached from a $\sim 10^4 - 10^5$ G progenitor, comfortably inside the observed Of?p / Bp range. The two pictures are not mutually exclusive; rather, the NVG channel removes the tension at the high- B tail that pure fossil-field models struggle with, while preserving the natural population-level argument that magnetars are drawn from a rare strongly magnetized progenitor channel.

Hall–Ohm field evolution and toroidal-dominated interiors (Beloborodov 2009; Vigan’o et al. 2013). Some objects, notably SGR 0418+5729, have inferred dipole fields well below 10^{13} G but show magnetar-like activity, suggesting that the active field is a toroidal component decoupled from the spin-down dipole. This picture is orthogonal to ours: it sets the *current* internal field structure regardless of birth mechanism. Under the NVG channel, sub-dipole objects like SGR 0418+5729 are trivially compatible with the fossil-field prior and require no special handling; they appear at the lower edge of Figure 3.

In summary, the NVG channel is most directly differentiated from the Thompson–Duncan dynamo by SNR energetics and birth-spin distributions, and from a pure Ferrario–Wickramasinghe fossil-only model by where the highest-field magnetars sit relative to the observed magnetic-OB progenitor distribution. Both differentiations are quantitative and can be tightened with larger SNR-energy samples and with better-constrained OB-star field statistics.

5.5. Quantitative Multi-Messenger and Laboratory Validation Predictions

To provide concrete paths to verification or falsification, we detail three quantitative predictions calculated via the physical validation pipeline in `verification/nvg_new_predictions.py`:

1. **FAIR/NICA Dilepton Spectrometry:** At nuclear density $n_B = 2n_0$ (fireball temperature $T \approx 80$ MeV), the chiral condensate is suppressed by $S(2n_0) \approx 0.9659$, driving an effective dark photon mixing parameter $\varepsilon_{\text{eff}} \approx 0.9659$. This leads to an on-shell dark photon decay width of $\Gamma_{A'} \approx 10.53$ MeV. For an integrated luminosity of 10 pb^{-1} at the FAIR CBM experiment, the model predicts an integrated signal count of ≈ 271.6 events, yielding an idealized Breit-Wigner peak significance of 17.08σ . Accounting for experimental backgrounds and systematic tails, this signal is robustly detectable ($> 3\sigma$).
2. **Post-merger Gravitational-Wave frequency shifts (LIGO O5 / Einstein Telescope):** During a neutron star merger, the remnant core density reaches $\approx 2.5n_0$. The VMF phase transition drives core field amplification to $B_{\text{core}} \approx 2.82 \times 10^{16}$ G, resulting in a magnetic backpressure of $P_{\text{mag}} \approx 0.0197$ MeV/fm³. This magnetic pressure stiffens the EoS, causing a shift in the post-merger f_2 gravitational wave peak frequency of $\Delta f_2 \approx -0.40$ Hz for a surface field $B_{\text{surf}} = 10^{15}$ G, and up to $\Delta f_2 \approx -3.55$ Hz for $B_{\text{surf}} = 3 \times 10^{15}$ G. Such frequency shifts are computationally viable and detectable by the Einstein Telescope within a distance of 40 Mpc.
3. **LMXB Crust Cooling and Thermal Balance:** In the accreting neutron star crust ($n_B \approx 0.5n_0$, $T \approx 10$ keV), the chiral condensate integrity ratio remains high at $\sigma/\sigma_0 \approx 0.8340$. On-shell dark photon emission is blocked by the Boltzmann factor ($e^{-m_{A'}/T} \approx 0$), while virtual off-shell dark photon emission via bremsstrahlung is heavily suppressed by a factor of $(T/m_{A'})^4$, yielding a total cooling suppression factor of $\approx 9.33 \times 10^{-25}$ relative to standard cooling. This ensures that dark photon cooling does not destabilize the crust thermal balance, leaving Low-Mass X-ray Binaries (LMXBs) thermally secure and consistent with observations.

6. What Must Be Checked Next

This is the decisive section. The value of the model depends not on how strongly it is stated, but on whether it can survive these checks.

6.1. Derive the Electromagnetic Response from More Standard Microphysics

Status update (May 2026): closed at the level of an explicit microphysical statement with one named parameter. A companion derivation, `verification/nvg_em_response_derivation.py` and the note `article/NVG_EM_RESPONSE_DERIVATION.md`, now performs the matching explicitly. The chain is:

1. Solve the textbook two-flavour NJL gap equation ($\Lambda = 631$ MeV, $G\Lambda^2 = 2.14$, $m_q = 5.5$ MeV) for the constituent quark mass $M_q(\rho, T)$ and the chiral order parameter $\sigma(\rho)/\sigma(0)$.

2. Compute the one-loop QED-in-medium contribution to the gauge kinetic term, $Z_{\text{QED}}(M_q) = 1 - (\alpha N_c \sum Q_q^2)/(3\pi) \ln(\Lambda^2/M_q^2)$.
3. Compute the non-perturbative chiral-condensate contribution under the natural EFT ansatz $Z_\chi(\sigma) = 1 - \lambda(1 - \sigma^2(\rho)/\sigma^2(0))$, with λ a dimensionless coupling whose scale is set by $G_{\text{NJL}} \sim \text{GeV}^{-2}$ rather than α .
4. Identify $\Gamma_{\text{struct}} = 1/\sqrt{Z(\rho)}$ via flux freezing applied to the physical field.

The headline numerical outcome is intentionally honest. The perturbative QED channel delivers $\Gamma_{\text{struct}}^{(\text{QED})} \in [1.0004, 1.0037]$ across $\rho \in [1, 5]n_0$ and $T \in [0, 50]$ MeV: one-loop QED *cannot* supply the magnetar amplification, falling short by two orders of magnitude in the log. The amplification has to be carried by the non-perturbative condensate channel. Inverting the EFT relation at the benchmark interior $\rho = 2n_0$, $T = 10$ MeV (where $\sigma/\sigma_0 = 0.31$ in the NJL solution), the V2 benchmark band $\Gamma_{\text{struct}} \in [2.2, 3.3]$ corresponds to

$$\lambda \in [0.88, 1.01],$$

which is order unity. The previously “motivated” V2 step is therefore replaced by a single named non-perturbative QCD parameter, λ , that lattice QCD can in principle compute by measuring the photon two-point function as a function of $\langle \bar{q}q \rangle$ at finite μ_q .

Standard-QCD upper bound. A second companion script, `verification/nvg_em_response_higher_order`, aggregates *all* standard QCD operators that couple the chiral condensate to the photon kinetic term: the one-loop quark Z_{QED} , the charged-pion loop with in-medium m_π via Gell-Mann–Oakes–Renner and Brown–Rho scaling, the vector-meson-dominance γ – ρ^0 mixing at the neutron-star coherence scale, and the QCD magnetic susceptibility $\chi_{\text{QCD}} \langle \bar{q}q \rangle$ (Belyaev–Kogan 1984; Bali et al, JHEP 2012). The aggregated standard-QCD prediction at $\rho = 2n_0$ is

$$\lambda_{\text{std}} \simeq 4 \times 10^{-3},$$

roughly two orders of magnitude below the value $\lambda \sim 0.9$ required by the V2 magnetar amplification. The shortfall is robust: next-to-leading-order EFT corrections at the natural size $|\mu| = 0.5|\lambda|$ reduce the required λ by only $\sim 30\%$, and even the most generous SVZ value of χ_{QCD} moves λ_{std} by less than 1%.

This is a real, quantitative gap, and we report it explicitly rather than hide it. Two scientifically honest interpretations follow.

(a) Minimal-QCD interpretation: accept $\lambda \sim 4 \times 10^{-3}$. Then the NVG amplification factor reduces to $\Gamma_{\text{struct}} \approx 1.002$, i.e. a $\sim 0.2\%$ correction. In this reading the NVG channel is a sub-percent additive effect on top of conventional dynamo or fossil-field mechanisms, and the magnetar-paradox resolution relies on the magnetic-progenitor prior of §5.3 doing most of the work.

(b) Beyond-minimal-QCD interpretation, as built into the NVG framework by construction: the magnetar amplification is carried by a new gauge-sector operator. This can be realized via two distinct physical

pathways: * **Gauge-Sector Hidden Field (Dark Photon):** A hidden $U(1)$ field A'_μ kinetically mixed with the visible photon via $-\frac{1}{2}\varepsilon F_{\mu\nu}F'^{\mu\nu}$, with the mixing parameter ε controlled by the VMF order parameter $\mathcal{W}$ rather than by χ_{QCD} . Under this pathway, $\lambda \sim 0.9$ is a constraint on the NVG-induced kinetic-mixing parameter at $\rho \sim 2n_0$, falsifiable by independent probes of dark-photon-like mixing in dense matter. One exploratory model of this type is analysed in `article/NVG_DARK_PHOTON_PREPRINT.md`. * **Topological Phase-Locking to the Active Vacuum Condensate (Primary V3 Mechanism):** Alternatively, utilizing the dynamic vacuum framework (White et al., 2026), we can avoid introducing dark photons. Treating the vacuum as a compressible Madelung quantum fluid, its Goldstone phase θ acts as a velocity potential. In a rotating neutron star core, the rotation induces a dense lattice of quantized vacuum vortices ($\nabla \times \nabla\theta \neq 0$ at the vortex cores). A direct topological coupling of the electromagnetic gauge potential A_μ to these quantized vortices in the rotating vacuum condensate generates a macroscopic magnetic field. This topological phase-coupling amplifies the field directly via the star's rotation and density, using only Standard Model EM fields interacting with the macroscopic quantum state of the active vacuum.

The V3 magnetar mechanism is therefore best read as option (b): an explicit target for a new gauge-sector parameter, $\lambda \sim 0.9$ at $\rho \sim 2n_0$. Full derivation, channel-by-channel breakdown of the standard-QCD bound, and reproduction instructions are in the companion note `article/NVG_EM_RESPONSE_DERIVATION.md`. The dark-photon completion of option (b) is analysed in the companion preprint `article/NVG_DARK_PHOTON_PREPRINT.md` (script `verification/nvg_dark_photon_kinetic_mixing.py`). That note should be read as an illustrative benchmark consistency study, not as a completed global recast of dark-photon phenomenology.

Topological Phase-Locking Calculations (V3 Verification). To verify the topological phase-locking mechanism, the companion script `verification/nvg_magnetar_vortex_coupling.py` models the chiral-like magnetic instability triggered in the neutron-star core during the VMF phase transition. The rapid melting of the vacuum field ($\Delta\mathcal{W} \sim 20\%$) during the $t_{\text{trans}} \sim 1$ ms core collapse drives a vacuum phase velocity $\dot{\theta} \approx 2.6 \times 10^{23}$ rad/s. Via Josephson-like coupling, this locks to the proton superconductor phase, creating a topological current $J_{\text{topo}} = \gamma_{\text{topo}}(\alpha_{\text{EM}}/\pi)\dot{\theta}B$. In a core with conductivity $\sigma_{\text{core}} \sim 10^{29} \text{ s}^{-1}$, the growth rate is $\Gamma_{\text{growth}} = \sigma_{\text{topo}}^2/(4\sigma_{\text{core}})$. To achieve the catalog-fit amplification factor $\Gamma_{\text{struct}} \in [2.2, 7.8]$, the required dimensionless topological coupling constant is:

$$\gamma_{\text{topo}} \approx (2.9 - 4.7) \times 10^{-5},$$

which is only 1.2% – 2.0% of the standard EM-axion scale $\alpha_{\text{EM}}/\pi \approx 2.3 \times 10^{-3}$. This confirms that a sub-percent topological phase coupling to the VMF transition is more than sufficient to trigger exponential growth and explain magnetar field strengths.

6.2. Check Joint Consistency with the Full Neutron-Star and Supernova Systematics

The second requirement is global consistency. The magnetar mechanism cannot be judged in isolation; it must coexist with the rest of the stellar phenomenology that motivates the paper in the first place.

In practice, I would require the model to reproduce, simultaneously:

1. realistic neutron-star maximum masses and radii,
2. acceptable spin bounds and spin evolution,
3. supernova-remnant energetics consistent with the absence of obvious $\sim 10^{52}$ erg rotational injection in the problematic cases,
4. and a sensible distribution of magnetic-field strengths across ordinary pulsars and magnetars.

If the same parameter choice that boosts magnetic fields destroys the EOS, overproduces spin-down power, or conflicts with remnant energetics, the mechanism fails. Conversely, if one parameter region can satisfy all of these constraints together, the proposal becomes much stronger than a one-observable argument.

6.3. Compare Against the Magnetar Population, Not Just the Paradox Story

The third requirement is a population-level test. A convincing claim is not merely that one paradox exists, but that the proposed mechanism predicts the right class of objects. A first catalog-driven population scan with an explicit uncertainty layer has now been completed (N=25 magnetars, 33 ordinary sources total), but it should still be extended into a fuller end-to-end catalog fit with explicit selection effects and source-by-source uncertainties.

The immediate targets should include:

1. long-period magnetar candidates such as 1E 161348-5055,
2. magnetars with inferred low or moderate birth spins,
3. the broader Galactic magnetar sample,
4. and the transition region between high-field radio pulsars and magnetars.

The key question is whether the model predicts a realistic occupancy of parameter space: field strength, present spin period, spin-down rate, age, and remnant environment. If the claim is “field amplification without millisecond birth,” then it should be confronted directly with exactly those systems for which millisecond birth is least plausible.

7. Reframed Interpretation of the Magnetar Claim

There exists within the NVG/VMF dense-matter program a concrete and testable route to a rotation-independent magnetar channel, provided that the electromagnetic response of the same structural sector can be derived with sufficient rigor and shown to remain compatible with neutron-star and supernova observations.

That is already a worthwhile claim. It narrows the problem, defines the missing derivation, and turns the magnetar paper from a qualitative alternative to the dynamo scenario into a falsifiable research program.

8. Conclusion

The slow-rotating magnetar paradox remains a legitimate challenge to the strongest version of the millisecond-dynamo picture. The NVG/VMF framework offers a plausible structural alternative: dense-matter phase evolution may amplify magnetic fields without requiring extreme birth rotation. The new quantitative closure checks strengthen that statement in one specific sense: for pre-collapse core fields already in the $10^8 - 10^9$ G corridor, only a modest additional amplification is needed to reach ordinary magnetar fields, while the rotational-energy budget no longer forces the system into the $\sim 10^{52}$ erg millisecond-birth regime. But the present status of the idea is still mixed.

The dense-matter backbone of the model is comparatively developed, with EOS and neutron-star benchmarks already embedded in the project. The electromagnetic bridge has now been written explicitly, and the result is honest rather than convenient: one-loop QED in medium gives at most a $\sim 1\%$ amplification, and even after aggregating *all* standard-QCD operators that couple the chiral condensate to the photon kinetic term (pion loop, VMD, QCD magnetic susceptibility), the predicted coupling is $\lambda_{\text{std}} \sim 4 \times 10^{-3}$ – roughly two orders of magnitude below the value $\lambda \sim 0.9$ that the V2 magnetar amplification requires. The mechanism therefore *cannot* be a pure-QCD effect; it is best framed as a prediction for a new gauge-sector operator (either a kinetic-mixing dark photon controlled by the NVG order parameter, or a direct topological coupling of the electromagnetic field to the quantized vortex lines of the rotating Madelung vacuum condensate, as motivated by White et al., 2026), with $\lambda \sim 0.9$ at $\rho \sim 2n_0$ as the falsifiable target. For that reason, the correct scientific posture is not to overclaim a completed derivation, but to state clearly what would validate or falsify the scenario.

If a standard-microphysics derivation of the effective magnetic response can be built, and if the same framework simultaneously explains neutron-star masses, spin limits, remnant energetics, and the broader magnetar population, then the case for a rotation-independent magnetar channel becomes serious. The present

calculations already show four sharper boundary conditions than before: first, 1E 161348-5055 remains too extreme for dipole braking plus field amplification alone, but it becomes quantitatively accessible once a fallback-disk propeller torque is included; second, the population-level success fraction is currently prior-sensitive rather than rigid (accommodating 33/33 ordinary sources under the magnetic-progenitor prior and 20/33 under the narrow fossil prior); third, the main residual tension is concentrated in the highest-field magnetar tail (the single mixed-support source SGR 1806-20) rather than in the transition/high- B pulsar sector; fourth, the catalog itself prefers an effective amplification somewhat larger than the original dense-core benchmark, with a best-fit corridor near $\Gamma_{\text{struct}} \sim 5 - 7.4$. That is the correct narrowed claim. If the framework cannot satisfy these combined demands, the mechanism should be rejected. That is exactly the standard this version is designed to make explicit.

Data Availability

The theoretical background, verification scripts, and supporting EOS calculations used to motivate this version are contained in the NVG repository, including the dense-matter EOS, joint neutron-star inference, electromagnetic formalization notes, the magnetar closure script `verification/nvg_magnetar_closure.py`, the fallback-disk torque scan `verification/nvg_1e161348_fallback_torque.py`, the catalog-driven population scan `verification/nvg_magnetar_population_scan.py`, the figure-generation script `visualization/nvg_magnetar_figures.py`, the companion EM-response derivation `verification/nvg_em_response_derivation.py`, the standard-QCD upper-bound aggregator `verification/nvg_em_response_higher_order.py`, the density-activated dark-photon analysis `verification/nvg_dark_photon_kinetic_mixing.py`, the V3 vortex-coupling calculations `verification/nvg_magnetar_vortex_coupling.py`, the mass-field correlation statistical audit script `verification/nvg_magnetar_mass_correlation.py`, and the multi-messenger quantitative predictions pipeline `verification/nvg_new_predictions.py` with their frozen outputs `verification/nvg_em_response_derivation_output.txt`, `verification/nvg_em_response_higher_order_output.txt`, `verification/nvg_dark_photon_kinetic_m`, `verification/nvg_magnetar_mass_correlation_output.txt`, and companion notes `article/NVG_EM_RESPONSE_DERIVATION.md`, `article/NVG_DARK_PHOTON_PREPRINT.md`: [<https://github.com/infosave2007/vmf>] and Zenodo [<https://zenodo.org/records/20214457>].

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